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St. Joseph's College of Arts & Science (Autonomous)

St. Joseph's College Road, Cuddalore–607001

B. Sc DEGREE EXAMINATION-APRIL 2021

BC611-HUMAN PHYSIOLOGY

Time: 3 hrs **Max Marks: 75**

SECTION-A (25X1=25)

Answer ALL the questions:

I. Choose the best answer (10X1=10)

1. The layer which is seen between the blood cells and the serum after centrifugation is _____.

   a) Buffy coat containing RBC b) Buffy coat containing WBC cells

   c) Buffy coat containing platelets d) Fat deposites

2. Which is the valve that is seen between the left atrium and left ventricle in human heart?

   a) Pulmonary valve b) Tricuspid c) Mitral valve d) Aortic valve

3. Which is called ptyalin?

   a) Pepsin b) Pancreatic Amylase

   c) Pancreatic lipase d) Salivary Amylase

4. Name the cell which produces HCL in the stomach?

   a) Parietal cells b) Oxyntic cells

   c) Delomorphous cells d) All of the above

5. The oxyhemoglobin dissociation curve describes _____.

   a) The relation between the partial pressure of oxygen and its saturation

   b) The oxygen saturation point

c) The relation between the partial pressure of carbondioxide and its saturation

d) The relation between the dissociation of oxygen to that of carbondioxide

6. Which hormone is essentially involved in the water balance from the kidney?

   a) Cortisol b) Vasopressin c) ADH d)Renin angiotensin

7. How many lobes are present in the human brain?

   a) 5 b) 4 c) 3 d) 6

8. Which one of the following neurotransmitter is commonly seen in the PNS?

   a) Acetylcholine b) Dopamine c) GABA d) Serotonin

9. Which is the structural and functional unit of the skeletal muscle system?

   a) Myosin filament b) Sarcomere

   c) Actin filament d) Both a and c

10. Which line demarcates the sarcomere in the microanatomy of muscle in the human system?

    a) I b) Z c) A d) M

II. Say true or false (5X1=5)

11. The outer covering of the human heart is endocardium.
12. The chemical digestion in the stomach by the HCl and pepsin forms chyme.
13. Glomerular capillaries contain fenestrae in their endothelia cells.
14. Gyri are the folds or ridges present in the cerebral cortex.

15. Cardiac muscles differ from skeletal muscles in that they contain intercalated discs.

III. Answer in one or two sentences: (10X1=10)

16. What is blood typing? Give example.
17. Define blood plasma.
18. List any two functions of bile acids.
19. What happens to the carbohydrates in the stomach?
20. What is chloride shift?
21. Define ultrafiltration in kidney.
22. What is reflex action?
23. What is cerebral meninx?
24. What is actin filament?
25. Which is bound to the myosin head?

SECTION-B (4X5=20)

Answer any FOUR of the following:

26. Comment on the composition of blood.
27. Give an account on Blood groups.
28. Give an account on the phases of gastric secretion.
29. With a suitable sketch explain the structure and function of nephron.
30. Explain synapse with a suitable diagram.
31. Explain the ultra-structure of skeletal muscle.

SECTION-C (3X10=30)

Answer any TWO of the following:

32. Describe the events occur during cardiac cycle in human heart.
33. Describe in detail digestion, absorption and assimilation of carbohydrates.
34. Explain how urine is formed.
35. With a suitable diagram explain the anatomy and function of cerebrum.
36. Describe in detail the mechanism of muscle contraction.

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